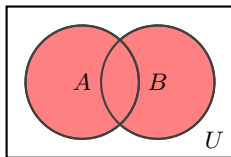


1. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ be the universal set, and $A = \{1, 2, 3, 4\}$ and $B = \{4, 6, 8\}$ be subsets of U . Find the indicated set, and draw a Venn diagram corresponding to the indicated set.

a. $A \cup B$

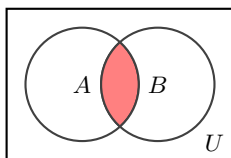
$$A \cup B = \{1, 2, 3, 4, 6, 8\}$$



The *Union*: elements in either A or B . Note that $A, B \subseteq A \cup B$. Do not repeat elements.

b. $A \cap B$

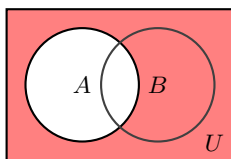
$$A \cap B = \{4\}$$



The *intersection*: elements in both A and B . Note that $A \cap B \subseteq A, B$. The intersection can be the empty set, $\emptyset$, e.g., $A \cap A' = \emptyset$.

c. A'

$$A' = \{5, 6, 7, 8\}$$

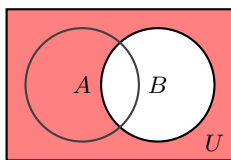


The *complement* of A : elements in U but not in A . Note that $A' \cup A = U$. In terms of set difference,

$$\begin{aligned} A' &= U - A \\ &= \{\cancel{1}, \cancel{2}, \cancel{3}, \cancel{4}, 5, 6, 7, 8\} = \{5, 6, 7, 8\} \end{aligned}$$

d. B'

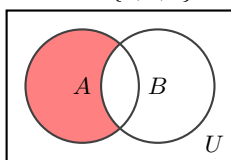
$$B' = \{1, 2, 3, 5, 7\}$$



$$\begin{aligned} B' &= U - B \\ &= \{1, 2, 3, \cancel{4}, 5, \cancel{6}, 7, \cancel{8}\} = \{1, 2, 3, 5, 7\} \end{aligned}$$

e. $A \cap B'$

$$A \cap B' = \{1, 2, 3\}$$



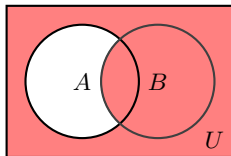
The intersection is just what both sets have in common,

$$\begin{aligned} B' &= \{1, 2, 3, 5, 7\} \\ A &= \{1, 2, 3, 4\} \\ A \cap B' &= \{1, 2, 3\} \end{aligned}$$

Equivalently, this is everything in A but not in B , so $A \cap B' = A - B$.

f. $A' \cup B$

$$A' \cup B = \{4, 5, 6, 7, 8\}$$

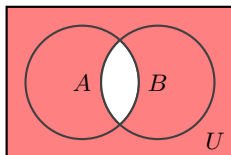


The union is just everything in both sets; don't repeat elements.

$$\begin{aligned} A' &= \{5, 6, 7, 8\} \\ B &= \{4, 6, 8\} \\ A' \cup B &= \{5, 6, 7, 8, 4, \cancel{6}, \cancel{8}\} = \{4, 5, 6, 7, 8\} \end{aligned}$$

g. $(A \cap B)'$

$$(A \cap B)' = \{1, 2, 3, 5, 6, 7, 8\}$$

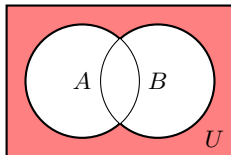


First find $A \cap B$. From above, $A \cap B = \{4\}$

$$\begin{aligned} (A \cap B)' &= U - (A \cap B) \\ &= \{1, 2, 3, \cancel{4}, 5, 6, 7, 8\} = \{1, 2, 3, 5, 6, 7, 8\} \end{aligned}$$

h. $(A \cup B)'$

$$(A \cup B)' = \{5, 7\}$$



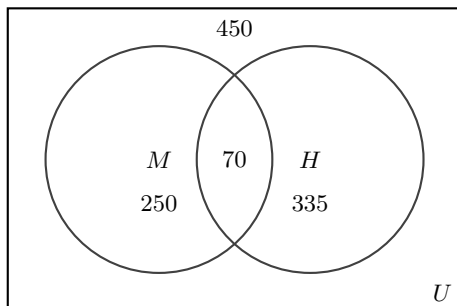
First find $A \cup B$. From above $A \cup B = \{1, 2, 3, 4, 6, 8\}$.

$$\begin{aligned} (A \cup B)' &= U - (A \cup B) \\ &= \{\cancel{1}, \cancel{2}, \cancel{3}, \cancel{4}, 5, \cancel{6}, 7, \cancel{8}\} = \{5, 7\} \end{aligned}$$

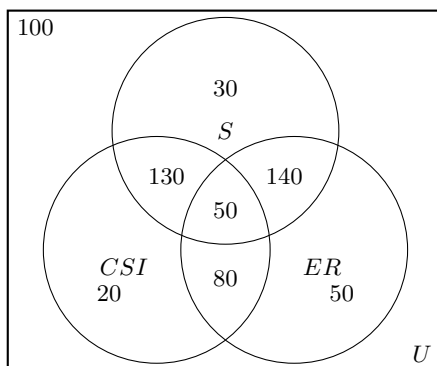
2. A college has 1105 students. This semester 320 students are taking a math course, 405 of the students are taking a history class, and 250 of the students taking a math class are not enrolled in a history class.

- Construct a Venn diagram, and write the appropriate numbers in the four regions
- How many of the students are taking math and history?
- How many students are taking history but not math?
- How many students are taking math or history?
- How many students are taking neither math nor history?

- (a) $n(M \cap H) = 320 - 250 = 70$, $n(H - M) = 405 - 70 = 335$, and $n((M \cup H)') = 1105 - 250 - 70 - 335 = 450$.



- (b) $n(H \cap H) = 70$
 (c) $n(H - M) = 335$
 (d) $n(M \cup H) = 250 + 70 + 335 = 655$
 (e) $n((M \cup H)') = n(U) - n(M \cup H) = 1105 - 655 = 450$
3. *Survivor*, *CSI*, and *ER* all belong to the Thursday night prime time lineup. A group of 600 college students were surveyed. 350 students watch *Survivor*, 280 watch *CSI*, and 320 watch *ER*. 50 students watch all three shows. 190 students watch both *Survivor* and *ER*. 180 students watch both *Survivor* and *CSI*. 50 students watch only *ER*.
- Construct a Venn diagram, and write the appropriate numbers in the eight regions.
 - How many students watch only *Survivor*?
 - How many students watch *CSI* or *Survivor*?
 - How many students watch *CSI* or *ER*?
 - How many students watch only *CSI*?
- (a) Number in *S* and *ER* but not *CSI* is: $190 - 50 = 140$, Number in *S* and *CSI* but not *ER* is: $180 - 50 = 130$. Number in *CSI* and *ER* but not *S* is: $320 - 50 - 140 - 50 = 80$. Number only in *S* is: $350 - 130 - 50 - 140 = 30$. Number only in *CSI* is: $280 - 130 - 50 - 80 = 20$.



- (b) From diagram: 30
 (c) From diagram: $30 + 130 + 140 + 50 + 80 + 20 = 450$, or by the union rule: $350 + 280 - 180 = 450$.
 (d) From diagram: $20 + 130 + 50 + 80 + 140 + 50 = 570$, or by the union rule: $280 + 320 - 130 = 470$
 (e) From diagram: 20
4. Find the value of each of the following:
- 4!
 $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$
 - 7!
 $7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$

c. $0!$

$$0! = 1$$

d. $\frac{10!}{7!}$

$$\frac{10!}{7!} = \frac{10 \cdot 9 \cdot 8 \cdot \cancel{7} \cdot \cancel{6} \cdot \cancel{5} \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot \cancel{1}}{\cancel{7} \cdot \cancel{6} \cdot \cancel{5} \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot \cancel{1}} = 720$$

e. $\frac{27!}{4!25!}$

$$\frac{27!}{4!25!} = \frac{27 \cdot 26}{4! \cdot 1} = \frac{702}{24} = 117. \text{ Similarly to above, everything before 25 cancels out.}$$

f. $\frac{1000!}{998!}$

$$\frac{1000!}{998!} = \frac{1000 \cdot 999}{1} = 999,000. \text{ Similarly to above, everything before 999 cancels out.}$$

g. ${}_7P_2$

$${}_7P_2 = \frac{7!}{(7-2)!} = \frac{7!}{5!} = \frac{5040}{120} = 42 \text{ or } \frac{7!}{5!} = 7 \cdot 6 = 42.$$

h. ${}_6C_4$

$${}_6C_4 = \frac{6!}{(6-4)!4!} = \frac{6!}{2!4!} = \frac{6 \cdot 5}{2!} = \frac{30}{2} = 15$$

5. A certain model of pickup truck is available in five exterior colors, three interior colors, and three interior styles. In addition, the transmission can be either manual or automatic, and the truck can have either two-wheel or four-wheel drive. How many different versions of the pickup truck can be ordered?

We have five categories, $_ \times _ \times _ \times _ \times _$. Filling in the number of choices for each category we have,

$$\underline{5} \times \underline{3} \times \underline{3} \times \underline{2} \times \underline{2} = 180$$

6. A certain fraternity must elect a president, vice president, and treasurer from its 30 members. How many ways can this be done?

There are no repeats and order does matter (since order will determine who is pres., vice pres., and treasurer). So we use permutations,

$${}_{30}P_3 = \frac{30!}{(30-3)!} = \frac{30!}{27!} = 30 \cdot 29 \cdot 28 = 24,360$$

7. The local branch of a national business chain has 27 employees. If they all shake each other's hand at the company picnic, at least how many handshakes occurred?

There are no repeats and order does NOT matter (Joe shaking hands with Betty is the same as Betty shaking hands with Joe). So we use combinations,

$${}_{27}C_2 = \frac{27!}{(27-2)!2!} = \frac{27!}{25!2!} = \frac{27 \cdot 26}{2} = 351$$

8. Using a standard deck of 52 cards,

- a. how many ways can five cards be drawn?
- b. how many five card hands can be dealt?

- (a) Here order matters, e.g., $\{7\heartsuit, K\spadesuit, 2\spadesuit, 3\clubsuit, 3\diamondsuit\}$ is a different way to draw the cards than $\{2\spadesuit, 3\clubsuit, 3\diamondsuit, 7\heartsuit, K\spadesuit\}$ since the order is different. So we use permutations,

$${}_{52}P_5 = \frac{52!}{(52-5)!} = \frac{52!}{47!} = 311,875,200$$

- (b) Here order does not matter, e.g., $\{7\heartsuit, K\spadesuit, 2\spadesuit, 3\clubsuit, 3\diamondsuit\}$ is the same hand as $\{2\spadesuit, 3\clubsuit, 3\diamondsuit, 7\heartsuit, K\spadesuit\}$. So we use combinations,

$${}_{52}C_5 = \frac{52!}{(52-5)!5!} = 2,598,960$$

9. A three-person committee must be chosen from a group of ten men and twelve women. How many committees are possible if it must consist of the following?

- a. Three men.
- b. Three women.
- c. One woman and two men.
- d. One man and two women.
- e. One man and two women, and one of these is selected as a spokesperson of the committee.

(a) No repeats and order doesn't matter $\Rightarrow$ combination. ${}_{10}C_3 = \frac{10!}{(10-3)!3!} = \frac{10 \cdot 9 \cdot 8}{3!} = \frac{720}{6} = 120$

(b) No repeats and order doesn't matter $\Rightarrow$ combination. ${}_{12}C_3 = \frac{12!}{(12-3)!3!} = \frac{12 \cdot 11 \cdot 10}{3!} = \frac{1320}{6} = 220$

(c) Here we have two categories: $_ \times _$. For the first categories we have 12 choices (equivalently, ${}_{12}C_1 = \frac{12!}{(12-1)!1!} = 12$), and for the second we have 10 choices (equivalently, ${}_{10}C_2 = \frac{10!}{(10-2)!2!} = \frac{10 \cdot 9}{2} = 45$).

So we have $12 \times 45 = 540$ possibilities.

(d) Similarly to above, we have two categories. For the first categories we have 10 choices, and for the second we have 12 choices (equivalently, ${}_{12}C_2 = \frac{12!}{(12-2)!2!} = \frac{12 \cdot 11}{2} = 66$). Then we select one of the three to be a spokesperson.

So we have $10 \times 66 \times 3 = 1980$ possibilities.

Equivalently, use a permutation where the order of 2 members does not matter, $(10 \times \frac{12!}{10!} \times 3)/2! = 1980$ (like counting the ways to arrange the letters in "SCHOOL"; the order of the "O"'s does not matter).

10. Each student at State University has a student ID number consisting of five digits (the first digit is nonzero, and the digits can be repeated) followed by two of the letters A , B , C , and D (letters cannot be repeated). How many different student numbers are possible?

We have five numbers $0 - 9$ which can be repeated, but the first is nonzero: $9 \times 10 \times 10 \times 10 \times 10 = 9 \cdot 10^4 = 90,000$ possibilities, and then two of four letters which cannot be repeated. It is implied that the order of the letters does matter since $10000AB$ and $10000BA$ would be two different IDs. So for the letters we have, ${}_4P_2 = \frac{4!}{(4-2)!} = \frac{24}{2} = 12$ possibilities. For a total of $9 \times 10^4 \times 12 = 1,080,000$ possible ID numbers.